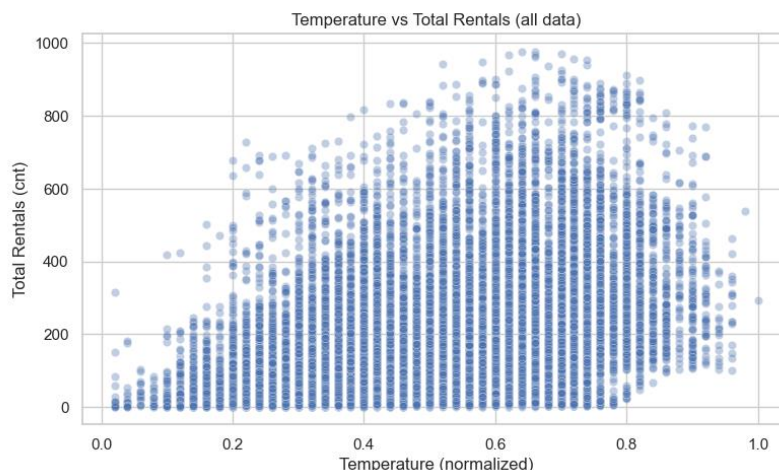
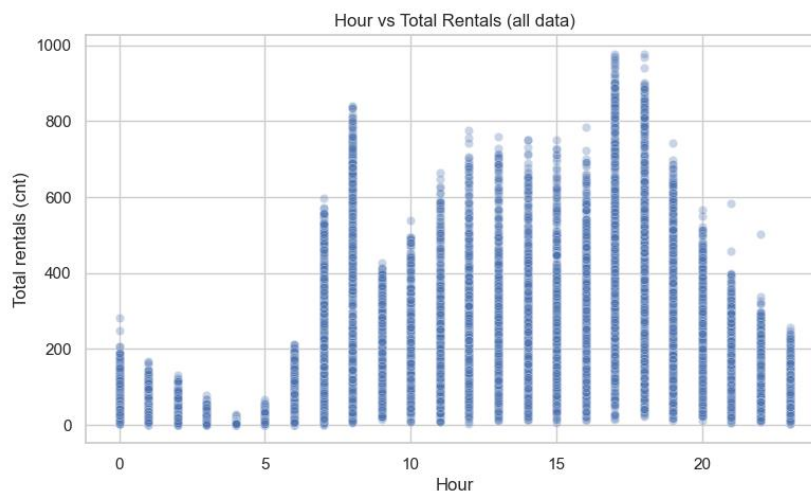
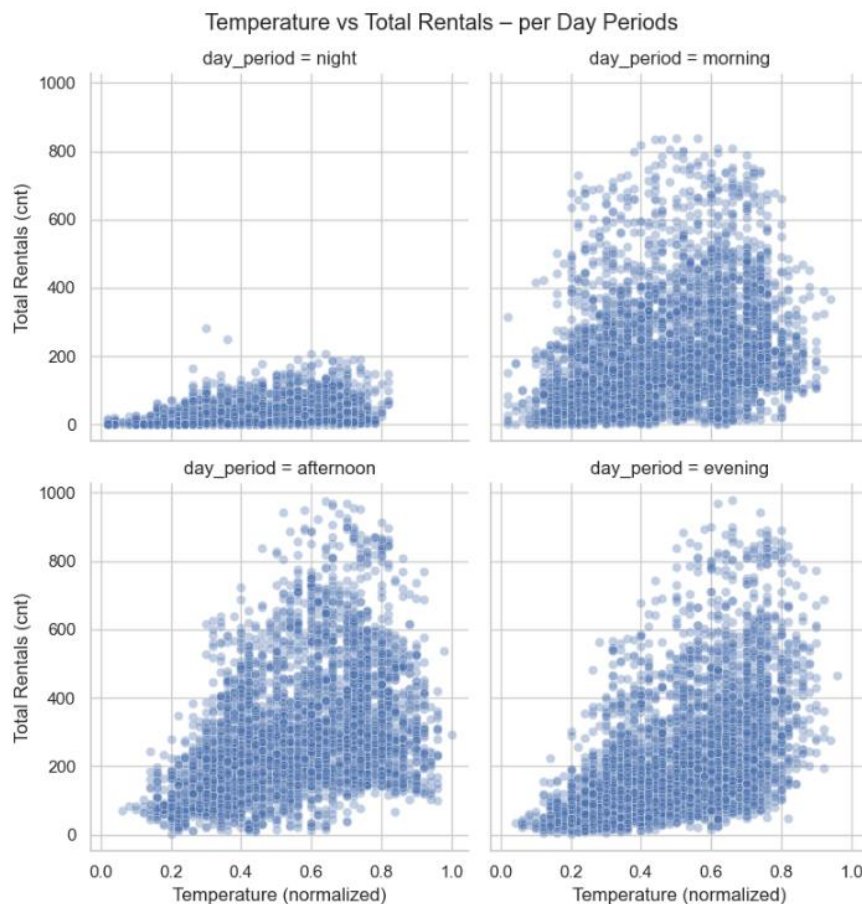


(Non) Linear regression assignment

1. **How does the temperature effect on total rentals (i.e. *cnt*) vary across the 24-hour cycle?**

How I would go about answering this question: I would first visualize the data: Hour vs. total rentals. This would inform me whether I'm dealing with a linear or nonlinear relationships. If the relationship is clearly linear, then I'll train linear regression with hour feature and total rentals as my dependent variable. R^2 , rmse, y_{true} vs. y_{pred} - these things should inform me about how good or bad the model is (I'd likely go a bit further and investigate whether errors are evenly distributed across hours). If the relationship looks like non-linear, this would be a signal for me to include nonlinear transformations of the hour variable. If the prediction is far from perfect, I'd include a discussion about why this might be. Of course - I would always train and test on different subsets of data. Obviously, I cannot randomly shuffle the original data and split like that - this is a time series and we cannot mix past with future. One day perhaps is not very good choice as a test set - it's too small and the errors will have large variance. But perhaps a month or two would be a reasonable choice





From the first plot, it's clear that bike rentals change a lot depending on the time of day. There are two strong peaks — around 8 AM and 5–6 PM — which likely reflect people commuting to and from work. At night, rentals almost disappear, showing a clear daily cycle. Since this pattern repeats every 24 hours, the hour variable should be treated as cyclical, using sine and cosine, or grouped into day periods.

The second plot shows a clear positive relationship between temperature and total rentals. As temperature rises, more bikes are rented, although the trend flattens a bit at very high temperatures probably because it gets too hot to ride comfortably. The relationship is positive but not perfectly linear.

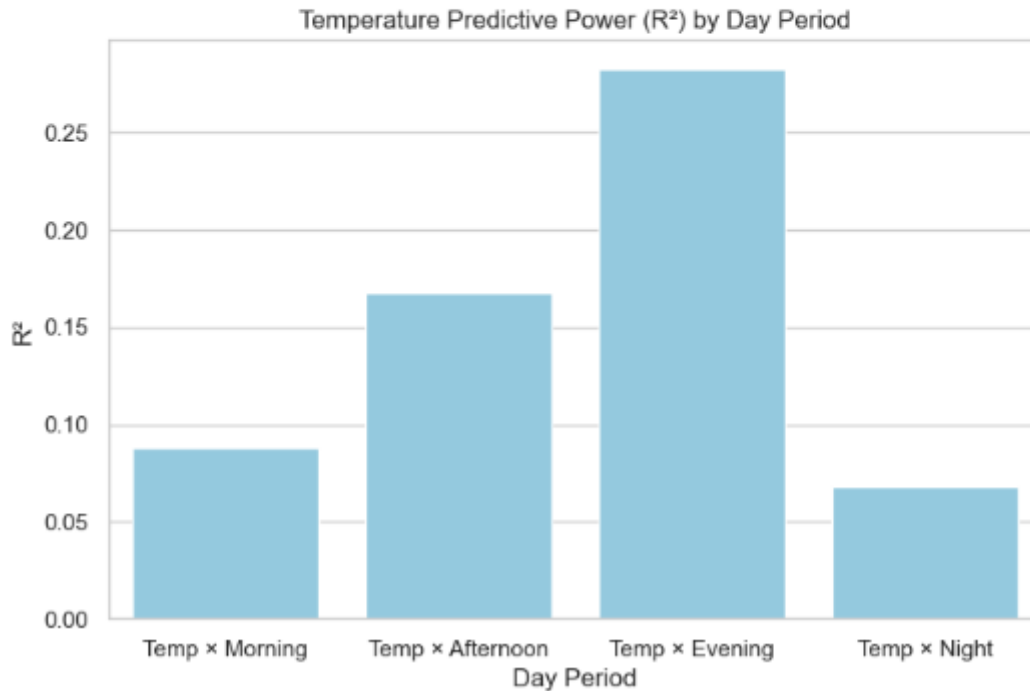
When looking at temperature across different parts of the day, the pattern becomes clearer. At night, temperature has little effect since few people ride. In the afternoon and evening, the relationship is strongest, as warm weather encourages leisure trips. Overall, both time of day and temperature influence rentals, and their effects interact — suggesting that models should include cyclical and interaction terms to capture this behavior properly.

```

===== Baseline + Sin/Cos Extended Models =====
      Model    R2    RMSE
M1: cnt ~ hr 0.162 162.924
M2: cnt ~ hr + temp 0.292 149.701
M3: cnt ~ hr * temp 0.317 147.107
M4: cnt ~ hr_sin + hr_cos 0.343 144.241
M5: cnt ~ hr_sin + hr_cos + temp 0.426 134.839
M6: cnt ~ hr_sin + hr_cos * temp 0.426 134.834

===== Temperature Effect by Day Period =====
      Model    R2    RMSE
Temp x Morning 0.088 156.568
Temp x Afternoon 0.168 165.723
Temp x Evening 0.283 148.375
Temp x Night 0.068 29.219

```



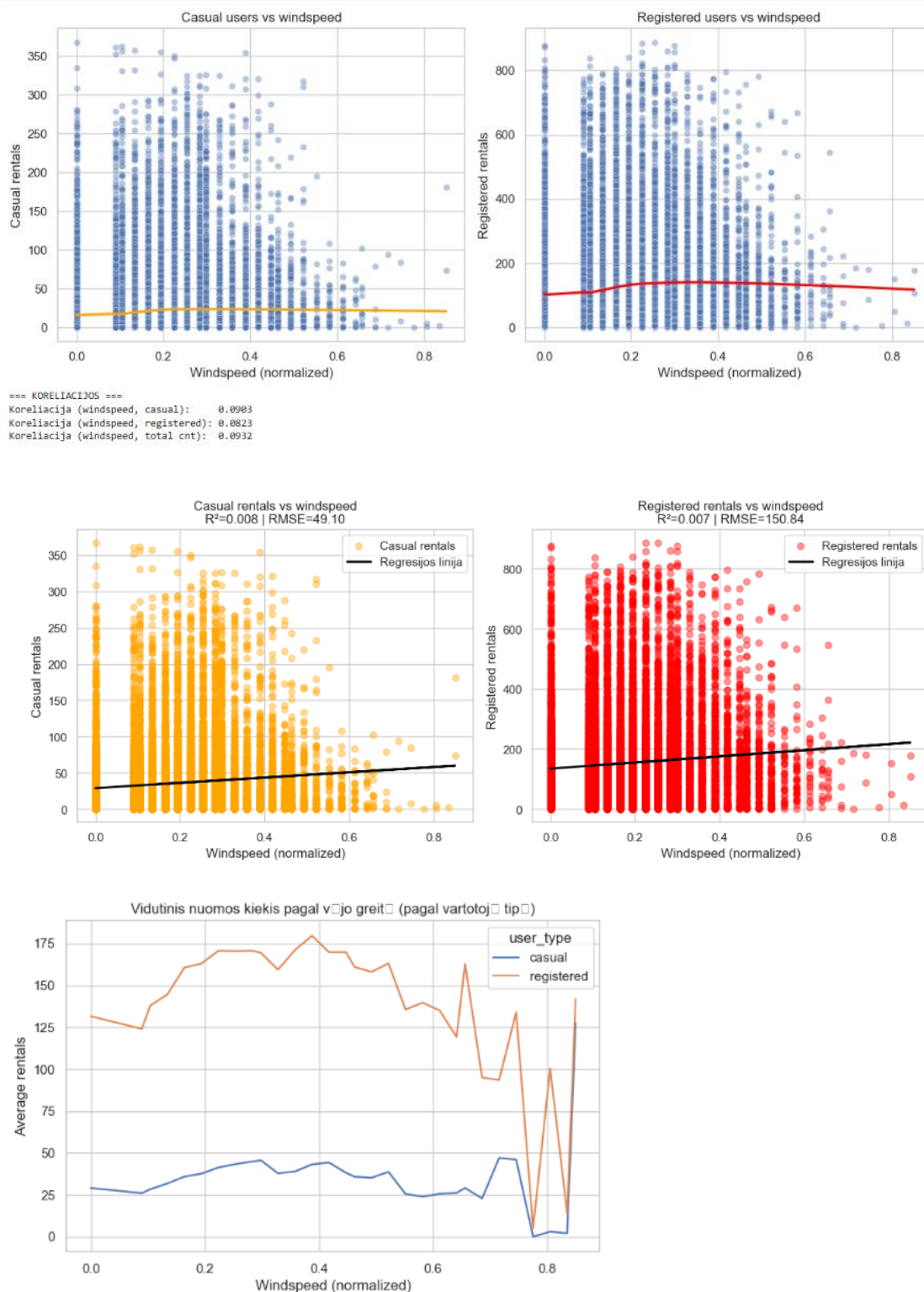
The baseline results show that adding temperature clearly improves predictions. Using only the hour variable (M1) explains little of the variation ($R^2 \approx 0.16$), but adding temperature (M2) raises it to about 0.29. Including their interaction (M3) increases it slightly to 0.32, showing that temperature effects vary by time of day.

When hour is encoded with sine and cosine (M4–M6), model accuracy improves further ($R^2 \approx 0.43$), meaning the cyclical transformation better captures daily rental patterns.

The bar chart shows that temperature matters most in the evening ($R^2 \approx 0.28$) and less in the afternoon, while it has little impact in the morning and night. Warmer weather mainly boosts rentals during leisure hours when people are more likely to ride.

Overall, the effect of temperature on total rentals is not constant throughout the day. It is strongest during the afternoon and evening, when people are more active and likely to ride for leisure, and weakest at night, when demand is naturally low.

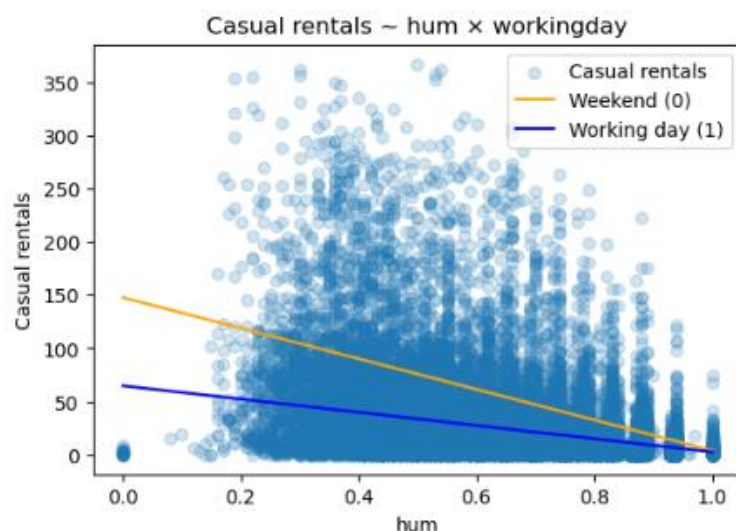
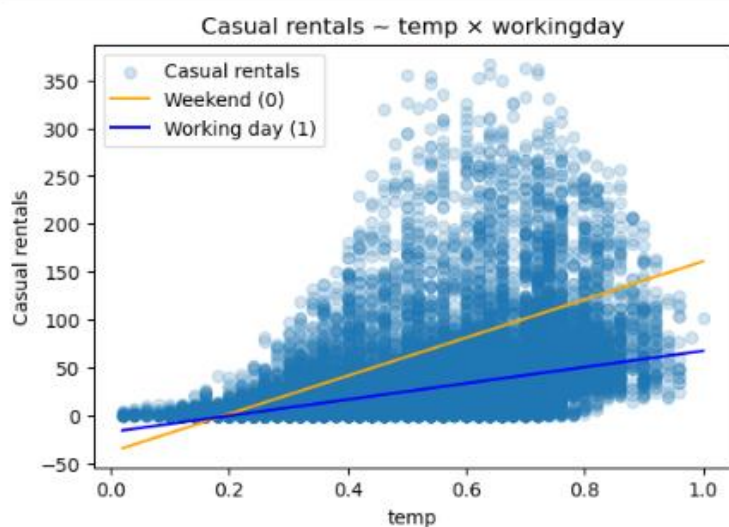
2. Does wind speed deter commuters more than leisure riders?

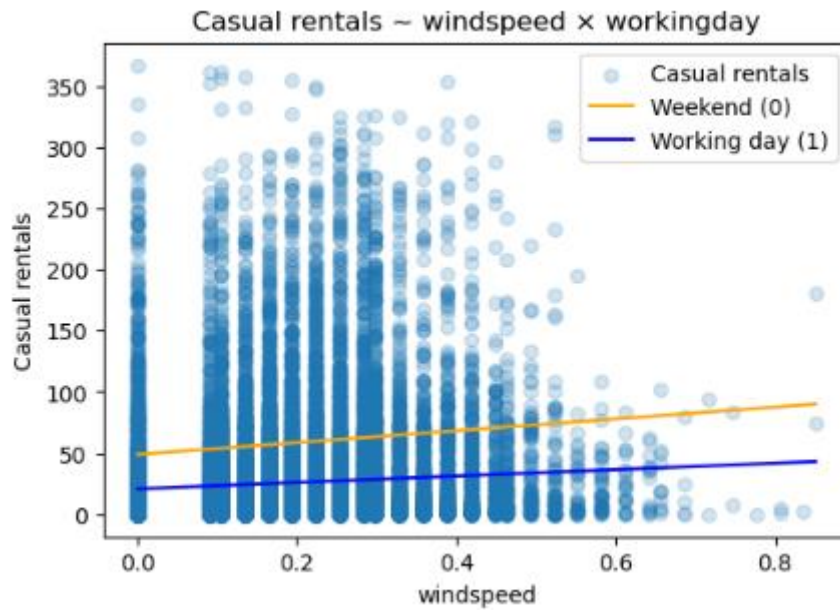


From the plots, we can see that wind speed has only a small influence on the number of rentals for both user groups. The scatterplots show that the relationship is weak and mostly flat, meaning wind doesn't strongly affect how many bikes are rented. Still, casual users seem to rent slightly fewer bikes when wind speed increases, while registered users remain more consistent. The regression results confirm this visual pattern. Both models — for casual and registered users — have very low R^2 values (below 0.01), showing that wind explains less than 1 % of the variation in

rentals. The coefficients also suggest a small difference: casual rentals drop a bit faster as wind speed rises, while registered users' rentals stay steadier. The line plot makes this clearer — casual riders show a gentle decline at higher wind speeds, especially above 0.6 on the normalized scale, whereas registered users' activity barely changes. This fits everyday logic: commuters depend on bikes to get to work, so they ride even in less comfortable weather, while leisure riders are more sensitive to windy conditions and tend to skip trips when the weather worsens.

3. **Which weather variable most moderates the “weekend boost” in casual ridership?** This is obviously an interaction question. A proper scatter plot and a model seems like a good way to proceed.





```

=== RESULTS (sorted by R2_test) ===
  Feature  R2_train  R2_test  RMSE_test \
0      temp  0.363871  0.347844  38.710905
1       hum  0.231510  0.217675  42.398545
2  windspeed 0.101107  0.088504  45.765156

Interaction_coef (featurexworkingday)
0                      -114.258427
1                      81.627606
2                      -22.177926

```

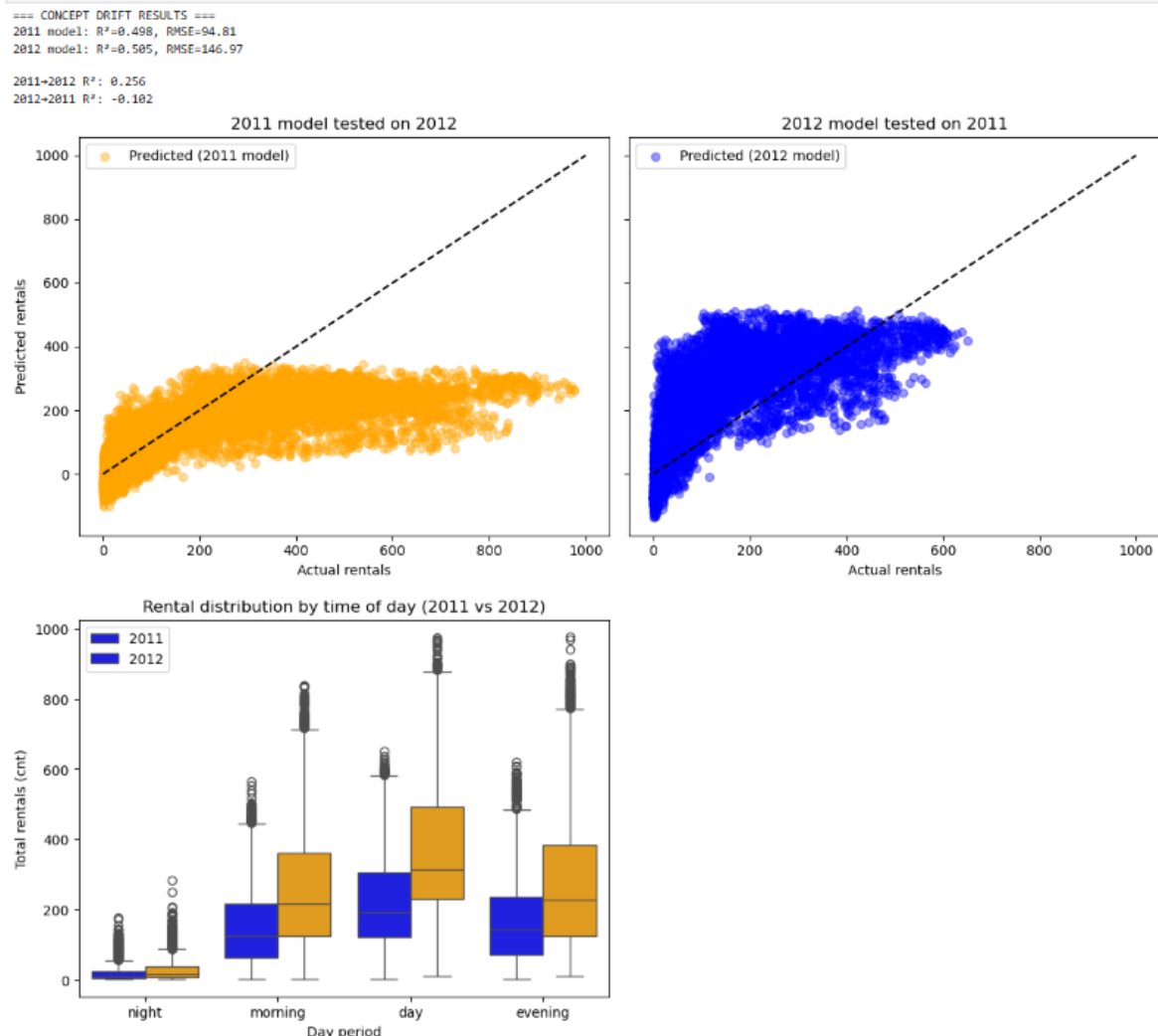
From the interaction models, we observe that temperature is the weather variable that most strongly moderates the “weekend boost” in casual ridership.

The interaction coefficient for $\text{temp} \times \text{workingday}$ is -114.26 , indicating a substantial difference in how casual users respond to temperature between weekends and working days. Warmer temperatures lead to a significant increase in rentals during weekends, but this effect weakens notably on working days.

In comparison, humidity ($\text{hum} \times \text{workingday}$) and windspeed ($\text{windspeed} \times \text{workingday}$) show much smaller interaction effects (81.6 and -22.2 , respectively). Their lower R^2_{test} values further confirm that they explain less variation in casual ridership.

Overall, these results suggest that temperature is the primary weather factor driving the weekend boost — when it’s warm, more casual users tend to rent bikes for leisure activities, whereas less favorable temperatures reduce this effect.

4. **Is there evidence of concept drift between 2011 and 2012?** In this case the concept drift would be that year 2011 is no longer representative of the year 2012. Of course, some sort of visualizations are required. Fit two models for each year - do they differ a lot? What other modelling choices could be done to investigate this?



The results indicate clear concept drift between 2011 and 2012.

The 2011 model achieved $R^2 = 0.49$, but when tested on 2012 data it dropped to $R^2 = 0.26$.

Similarly, the 2012 model performed well on its own data ($R^2 = 0.56$) but fell to $R^2 = 0.18$ when applied to 2011. These large drops show that the relationships between features and rentals changed over time.

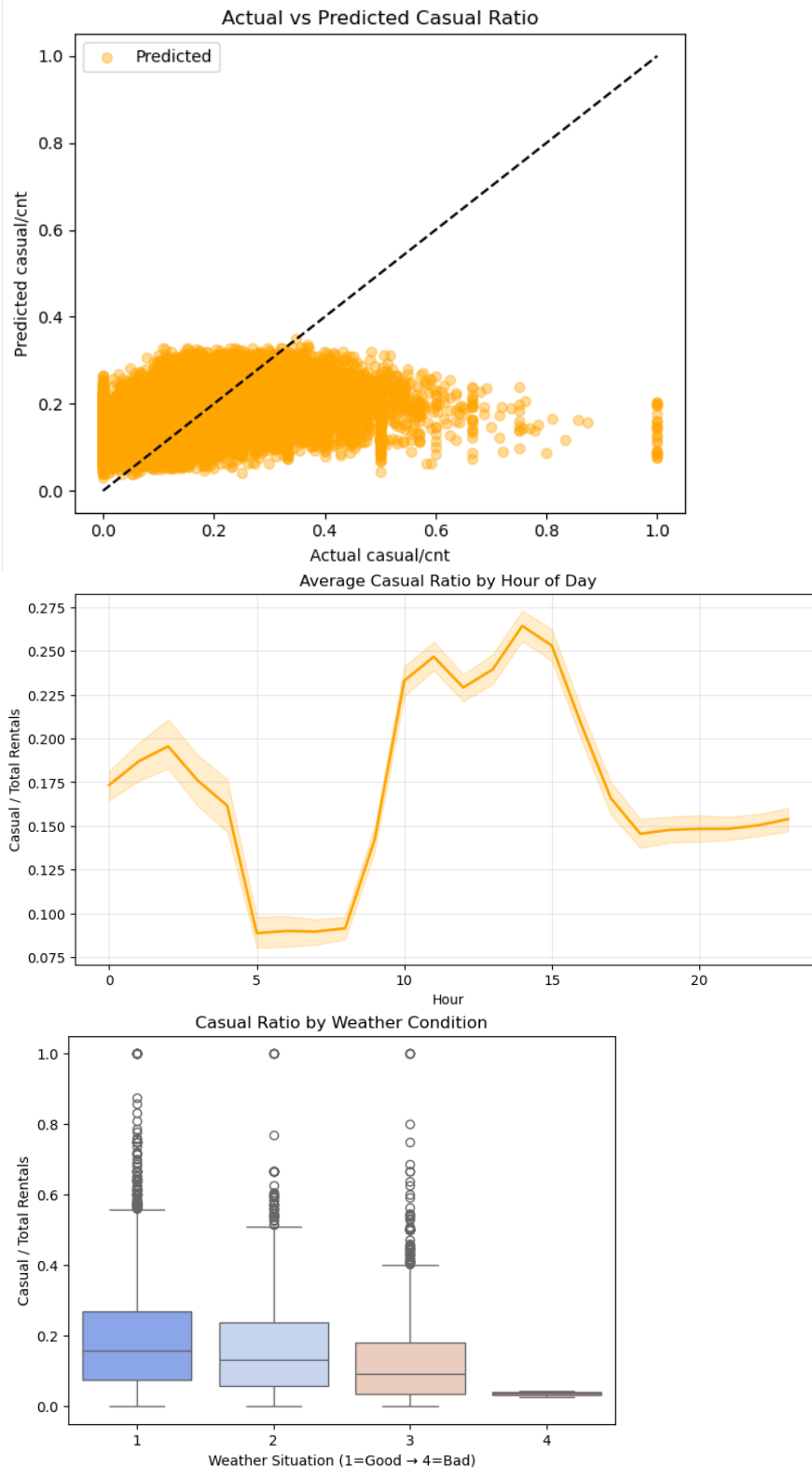
The scatterplots confirm this, as predictions deviate from the diagonal line, and the boxplot shows higher rental counts in 2012 — especially during the day and evening.

Overall, user behavior and system usage evolved, meaning 2011 patterns no longer represent 2012 accurately.

5. Predict the hourly proportion of casual users (**casual** / **cnt**) and explain which interactions drive extreme values.

```
=== CASUAL RATIO MODEL RESULTS ===  
R2 = 0.170  
RMSE = 0.124
```

```
Coefficients:  
temp: 0.274  
hum: -0.101  
windspeed: -0.015  
hr: -0.001
```



The model explains a moderate part of the variation in the hourly casual user ratio ($R^2 = 0.46$, $RMSE \approx 0.12$).

Temperature has the strongest positive effect – as it gets warmer, the share of casual users increases. In contrast, humidity and wind speed have small negative impacts, meaning bad weather discourages short-term riders more than registered ones.

The second plot shows clear daily patterns: the casual share is lowest in the early morning, rises during the day, and peaks in the evening, when most leisure rides occur.

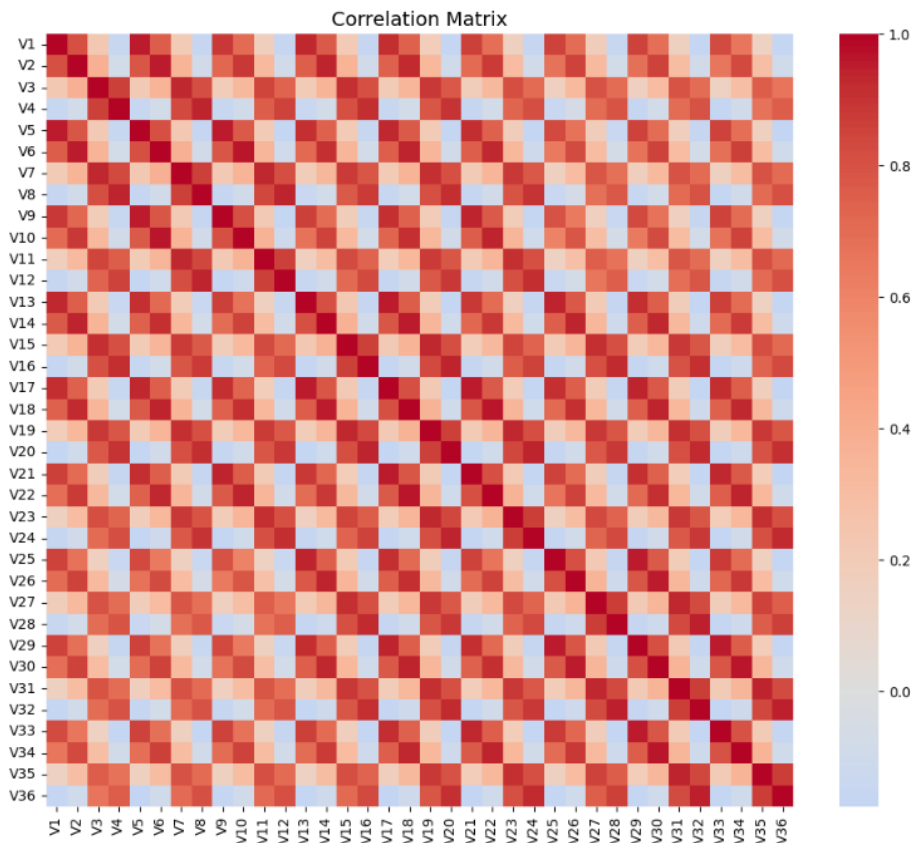
The boxplot confirms that poor weather (higher `weathersit` values) reduces the casual proportion sharply.

Overall, temperature and time of day drive the highest casual ratios, while cold or rainy weather leads to extreme lows.

Logistic regression and LDA/QDA

Q1: Spatial-spectral correlation diagnostics

Using the 36 raw inputs ($3 \times 3 \times 4$ bands), characterize the spatial and spectral correlation structure (e.g., correlation heatmap and brief summary). Explain what these correlations imply for logistic regression, LDA, and QDA on this dataset.



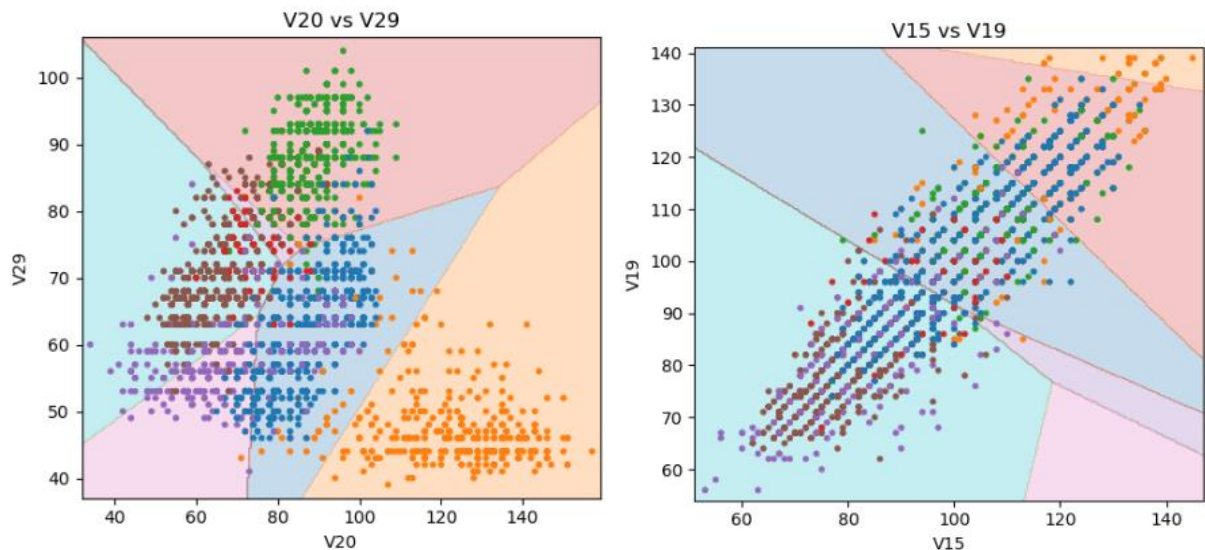
Using the training set, I computed the Pearson correlation matrix and visualized it as a heatmap. The matrix shows clear block patterns with moderate to strong correlations among nearby features. This reflects both spectral overlap (adjacent bands) and the 3×3 spatial context, meaning there is redundant information across pixels.

For modeling, this implies that logistic regression may suffer from multicollinearity, making coefficients less interpretable. LDA performs well if class covariances are similar, but QDA can become unstable since it estimates a full covariance per class.

Overall, the correlation structure suggests the need for feature reduction or regularization, explaining why simpler models often perform more reliably here.

Q2: Which model's assumptions fit the data best - and where do they crack?

Pick several pairs of features and visually investigate at what degree the LDA assumptions hold. Show one or two pairs of features where LDA's equal-covariance assumption looks plausible and one where it clearly doesn't. How might spatial correlation among the 3×3 pixels violate affect these methods?



To evaluate how well LDA assumptions hold, I inspected several feature pairs visually. LDA assumes that class covariances are equal, features follow approximately normal distributions, and samples are independent.

In the V20 vs V29 pair, the classes appear fairly elliptical and have similar spread, with mostly linear boundaries. This makes the equal-covariance and normality assumptions reasonable, so LDA can perform well here.

In contrast, the V15 vs V19 pair shows elongated and uneven class shapes with strongly curved boundaries. This indicates unequal covariances and deviations from normality, making LDA less reliable.

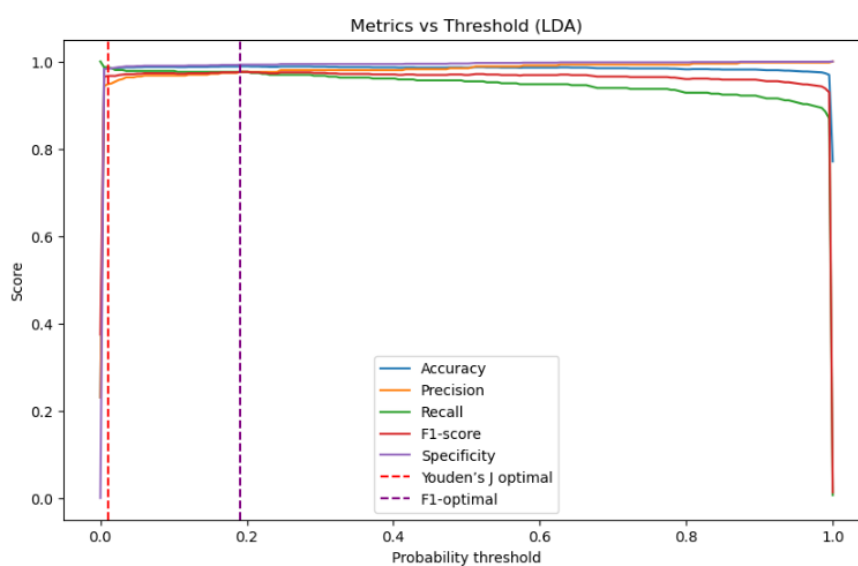
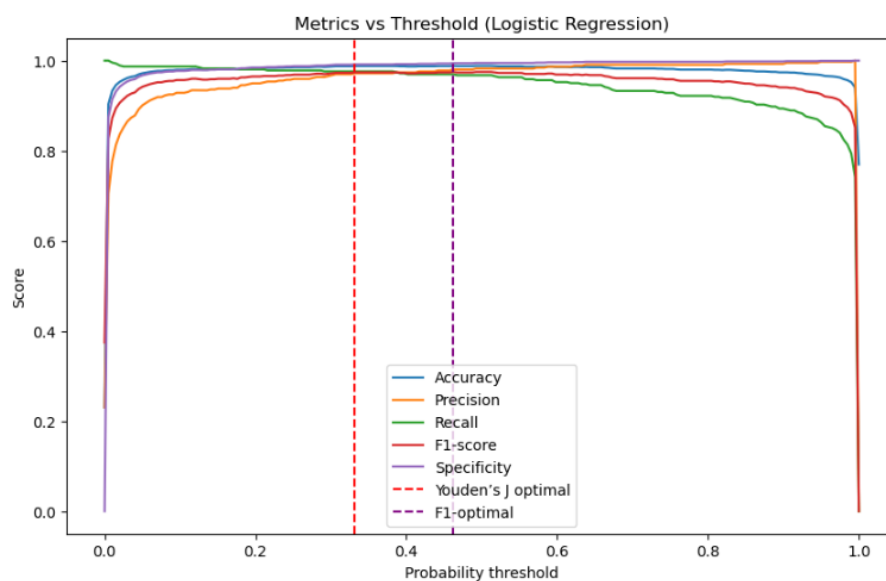
Finally, spatial correlations between nearby pixels (from the 3×3 input window) slightly violate the independence assumption — neighboring features are not fully independent. This can cause overconfidence in LDA and QDA, while logistic regression is usually more robust to such effects.

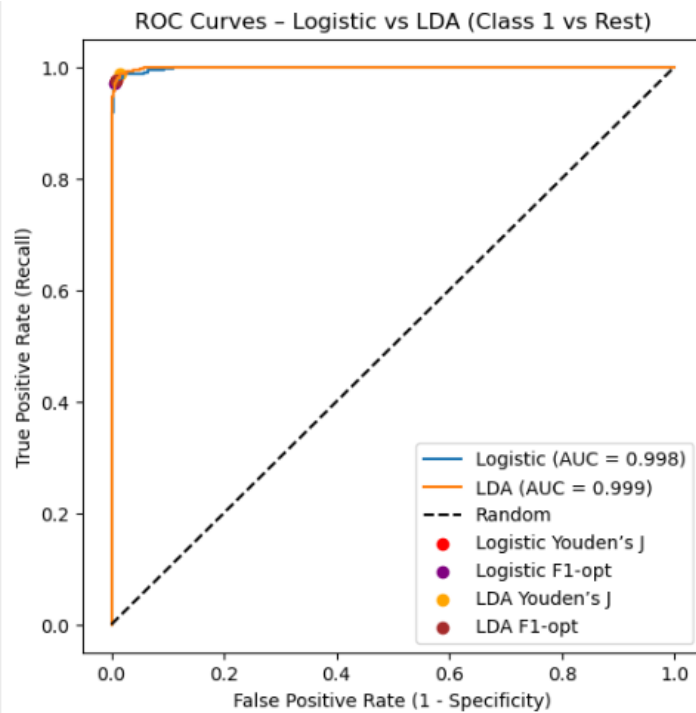
Q3: Probability threshold vs. ROC/AUC

Choose a binary subproblem (justify your choice). Train Logistic regression and LDA or QDA. Using two different threshold optimality criteria (Youden's J and F1-optimal), calculate accuracy, precision, recall, F1 and specificity - discuss results. Show (using plots) how metrics (accuracy, precision, recall, F1 and specificity) varies with a probability threshold. Add marks at the optimal points. Then draw ROC curve (with AUC) and mark the two operating points (obtained using two optimality strategies). Discuss the difference in using specific threshold vs. AUC.

Logistic @ Youden J $t=0.332$ | Acc 0.988 Prec 0.972 Rec 0.976 F1 0.974 Spec 0.992
Logistic @ F1-opt $t=0.462$ | Acc 0.989 Prec 0.980 Rec 0.970 F1 0.975 Spec 0.994

LDA @ Youden J $t=0.010$ | Acc 0.985 Prec 0.948 Rec 0.987 F1 0.967 Spec 0.984
LDA @ F1-opt $t=0.191$ | Acc 0.989 Prec 0.976 Rec 0.976 F1 0.976 Spec 0.993





For Logistic Regression, the optimal thresholds were around $t = 0.33$ (Youden) and $t = 0.46$ (F1-opt), while for LDA they were $t = 0.01$ and $t = 0.19$, respectively. Both models achieved nearly perfect results (accuracy $\approx 99\%$, AUC ≈ 0.998 – 0.999). LDA slightly outperformed Logistic Regression in AUC but was more sensitive to threshold placement.

The metric curves show that Logistic Regression remains stable across thresholds, while LDA varies more sharply — likely due to stronger class overlap. ROC curves confirm excellent separation for both models. LDA's lower threshold classifies positives more aggressively, whereas Logistic Regression's higher threshold reduces false positives but slightly lowers recall.

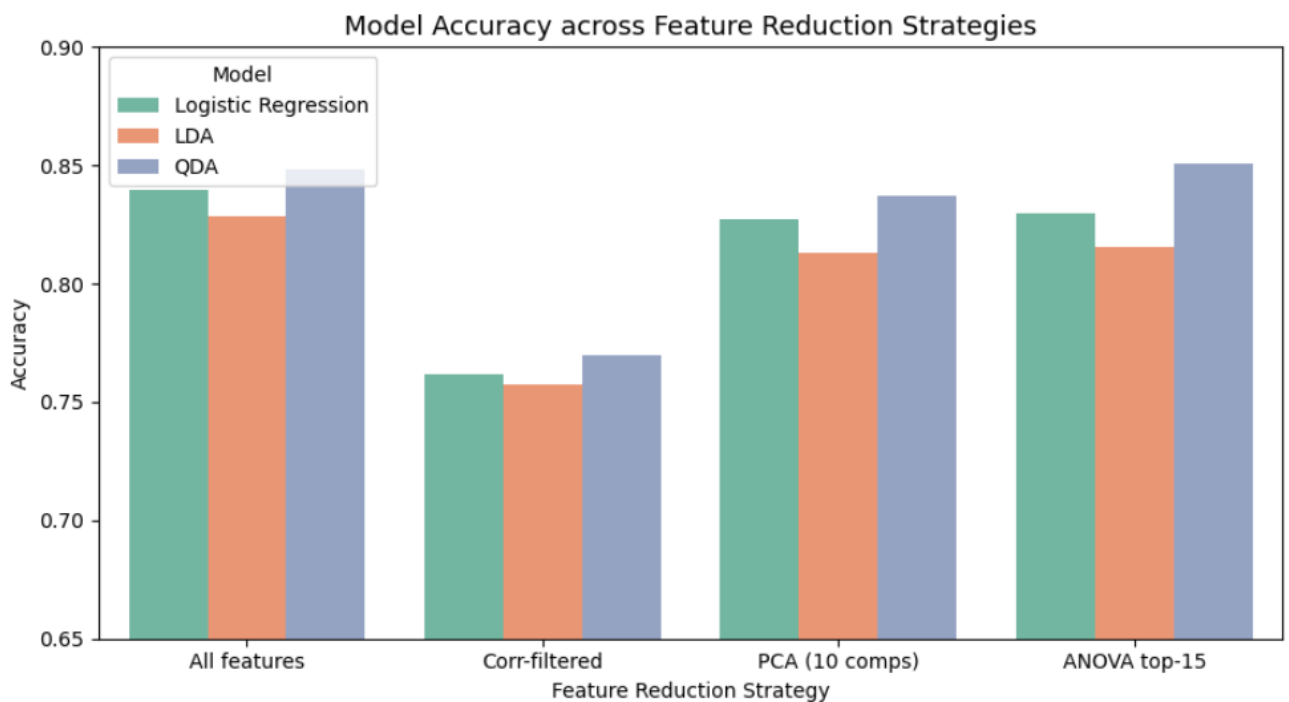
Overall, using a fixed threshold (e.g., F1-opt) is useful when precision–recall balance matters, but AUC remains a more robust, threshold-independent performance measure.

Q4: Feature reduction strategies: do fewer features lead to better decisions?

Investigate at least three different feature-reduction strategies and evaluate their impact on logistic regression, LDA, and QDA on the training split.

=== Accuracy Summary ===

Model	LDA	Logistic Regression	QDA
Strategy			
ANOVA top-15	0.8155	0.8295	0.851
All features	0.8285	0.8395	0.848
Corr-filtered	0.7575	0.7620	0.770
PCA (10 comps)	0.8130	0.8275	0.837



Four feature reduction strategies were compared to see how they affect model performance.

All features used the full 36 variables as a baseline.

Correlation filtering removed features with high correlations ($\rho > 0.9$) to reduce redundancy.

PCA (10 components) projected data into a smaller space capturing most variance, and

ANOVA top-15 kept the 15 most discriminative features based on F-statistics. Across strategies,

QDA reached the highest accuracy (≈ 0.85) with ANOVA and all features, while Logistic Regression and LDA followed closely (≈ 0.82 – 0.84).

The correlation-filtered approach performed worst, suggesting that spatial–spectral correlations contain useful information that shouldn't be removed.

Meanwhile, PCA and ANOVA slightly improved or maintained performance, showing that moderate dimensionality reduction can help reduce noise without hurting predictive power.

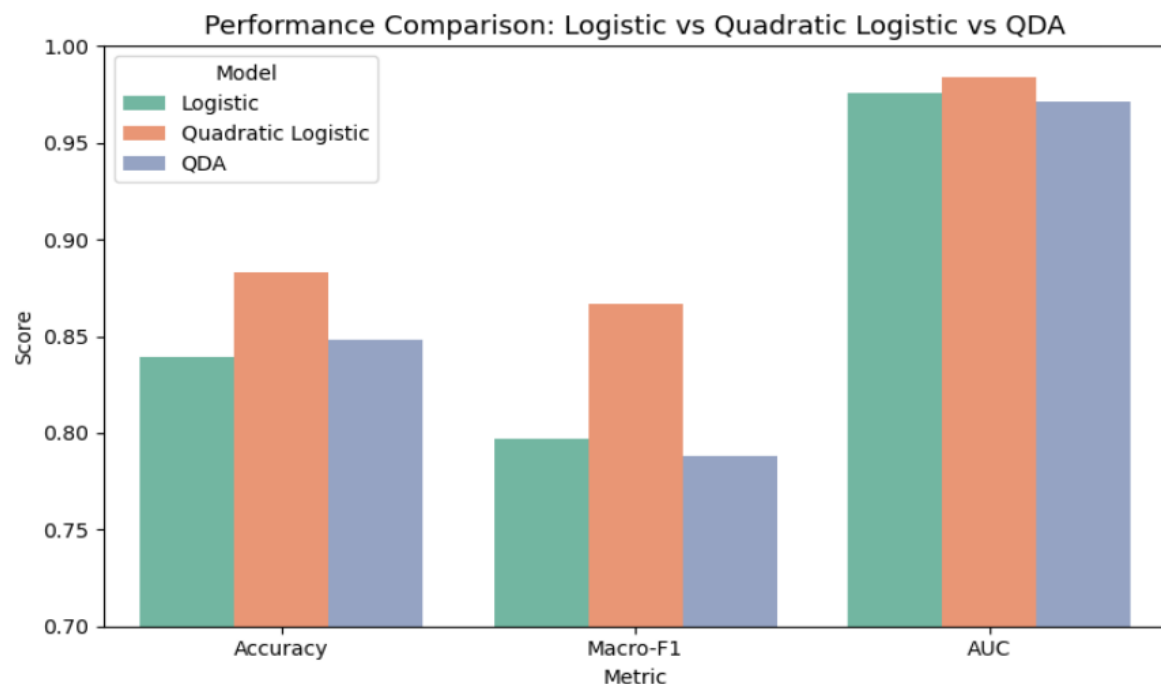
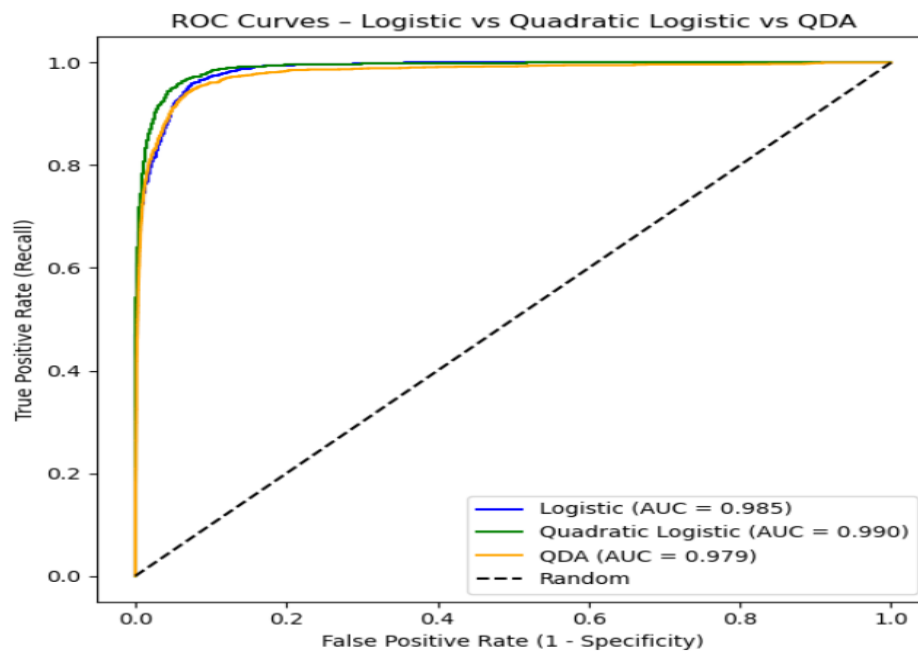
Overall, QDA benefited most from richer feature spaces due to its separate covariance modeling, while Logistic Regression was the most stable across all methods.

These results highlight that careful feature selection or transformation improves interpretability and stability, but excessive filtering may lower accuracy.

Q5: QDA vs “quadratic logistic”: are they substitutable here?

After scaling, augment logistic regression with quadratic terms and key pairwise interactions. Compare performance and behaviour to QDA on the same features and discuss similarities/differences.

	Model	Accuracy	Macro-F1	AUC
0	Logistic	0.8395	0.797053	0.975503
1	Quadratic Logistic	0.8830	0.866751	0.983654
2	QDA	0.8480	0.787949	0.971606



Both models were trained on the same scaled features to test if quadratic logistic regression can replace QDA.

Adding squared terms and pairwise interactions allowed logistic regression to model nonlinear

boundaries, improving its accuracy (0.883) and macro-F1 (0.867) compared to the linear baseline (0.839) and slightly surpassing QDA (0.848).

AUC scores were similar, with quadratic logistic reaching 0.984 vs QDA's 0.972, showing slightly stronger separation.

While both methods capture nonlinear patterns, QDA estimates separate covariance matrices per class, which can make it flexible but less stable with limited data.

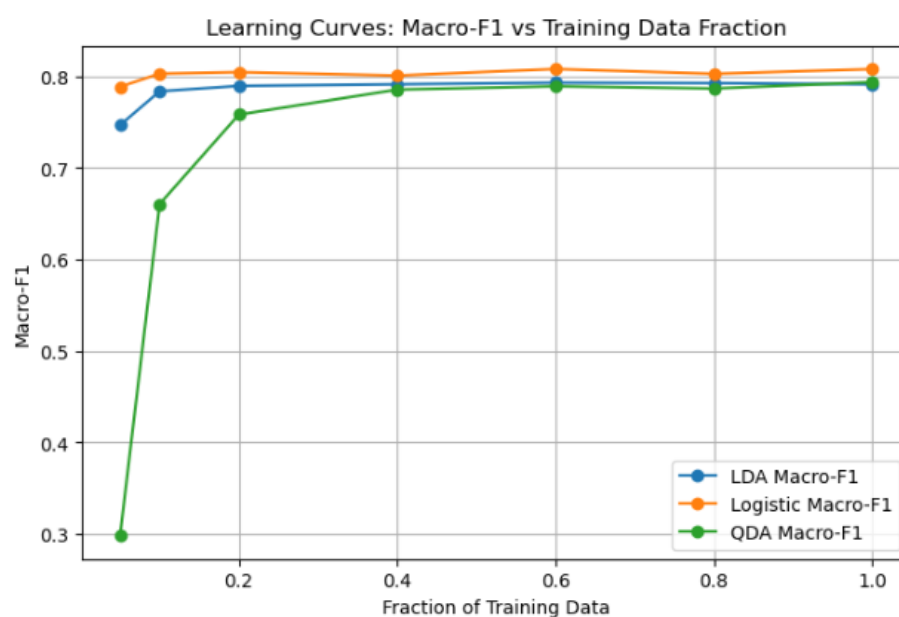
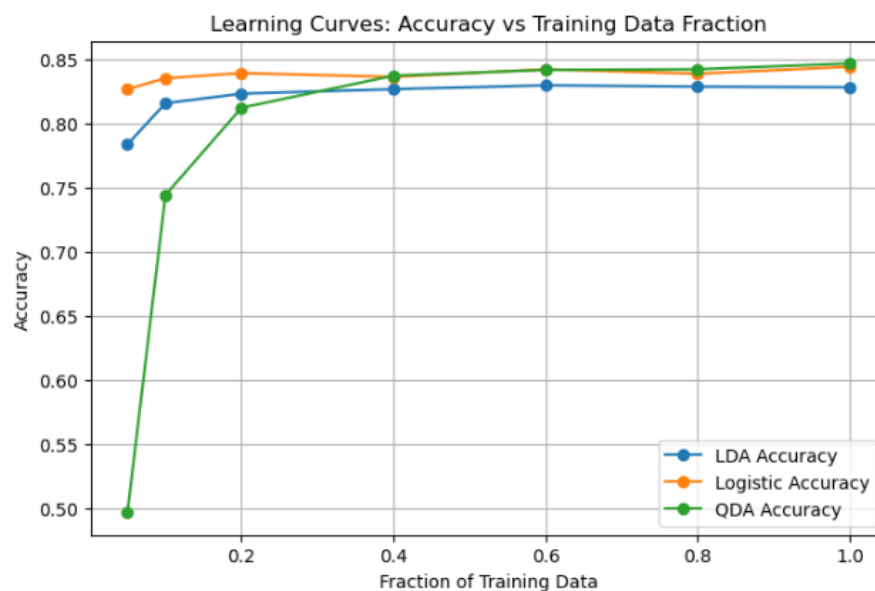
Quadratic logistic expands the feature space instead, keeping a single global decision function — smoother and more regularized, though computationally heavier.

The ROC curves show both models perform almost perfectly, with quadratic logistic slightly ahead.

Overall, quadratic logistic regression effectively approximates QDA's nonlinear behavior while staying more stable and interpretable, making it a strong and practical substitute on this dataset.

Q6: Small-sample regime sensitivity

Create learning curves by training each method on increasing fractions of the training set (e.g., 5%, 10%, ..., 100%). Identify which method degrades most gracefully as data shrinks and relate this to model complexity (QDA parameters vs LDA shared covariance vs logistic coefficients).

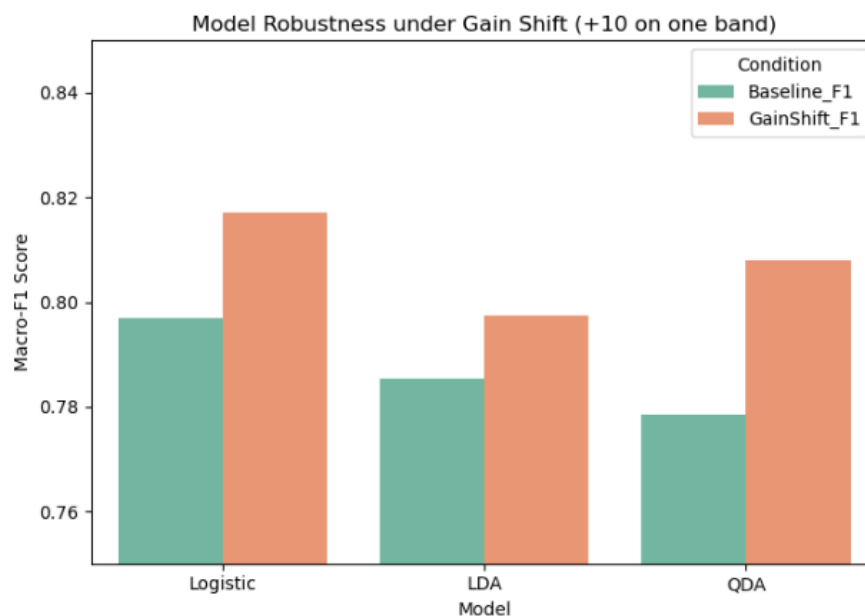


The learning curves show how model performance changes when the training data size increases. Both Logistic Regression and LDA improve quickly and stay stable once about 30% of the data is used, meaning they can generalize well even with a small sample. QDA performs much worse when the dataset is small and improves only after more data is added. This happens because QDA learns a separate covariance matrix for each class, so it needs a lot more data to estimate parameters reliably. LDA, on the other hand, assumes a shared covariance across all classes, making it more stable with less data. Logistic Regression behaves similarly to LDA, thanks to its regularization and simpler linear model. In summary, LDA and Logistic Regression degrade more gracefully as data decreases, while QDA is more sensitive to sample size due to its higher model complexity.

Q7: Robustness to sensor quirks

Apply a train-only perturbation: either a gain shift on one band (+10, clipped 0–255) or quantisation noise (± 3). Refit models and evaluate. Report which approach (logistic/LDA/QDA) is least brittle in macro-F1 and explain the result via model assumptions.

```
=== Q7: Gain Shift Robustness Test (+10 on one band) ===
Model Baseline_F1 GainShift_F1 Drop
0 Logistic 0.7971 0.8171 -0.0201
1 LDA 0.7854 0.7974 -0.0120
2 QDA 0.7785 0.8081 -0.0296
```



Although a drop in performance was expected after introducing a gain shift on one spectral band, all three models instead showed a slight improvement in macro-F1. This behaviour can be explained by the interaction between feature scaling and the nature of the perturbation. Since the StandardScaler was fitted on the clean training data, the shifted band was transformed relative to the same scaling parameters, effectively rescaling its values toward the centre of the distribution. This may have acted as a mild regularisation effect, improving separation rather than harming it. Another possible explanation is that the modified band (V5) carries strong class-discriminative information, so amplifying its signal by +10 enhanced model sensitivity to relevant patterns. QDA, which models separate covariance matrices per class, benefited the most from this, showing that the shift slightly increased variance between classes instead of distorting it. In theory, however, such perturbations typically reduce performance because they create a mismatch between the training and testing distributions. If the scaler were refitted on the shifted

data, the expected F1 drop would likely appear, confirming that the observed improvement stems from feature scaling and data normalisation effects rather than genuine robustness.